

Kernel Hotel Booking System

LOGICAL DATABASE DESIGN & ENTITY RELATIONSHIP MODEL

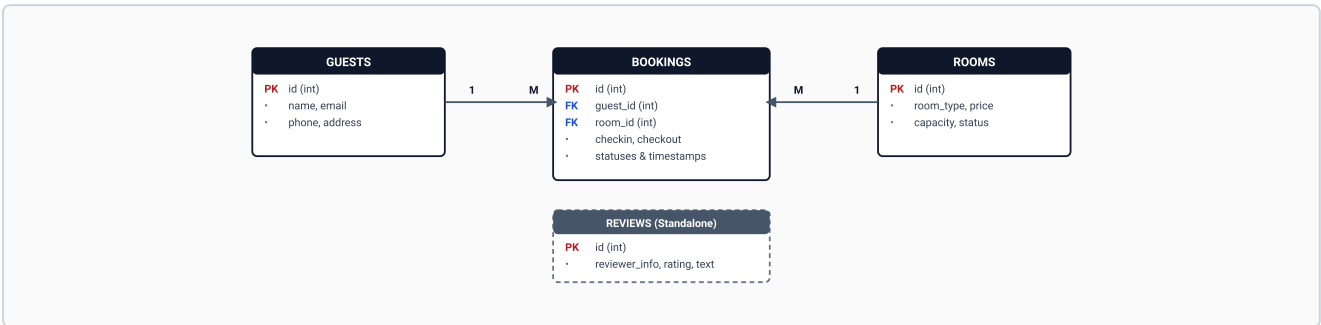
1. SYSTEM OVERVIEW

This document specifies the logical database architecture for the Kernel Hotel reservation system. The data model provides structural support for core hotel operations: managing guest accounts, maintaining real-time room inventory, handling reservations and transactions, and collecting guest reviews.

2. HIGH-LEVEL ENTITY SUMMARY

TABLE	PRIMARY KEY	FOREIGN KEYS	DESCRIPTION & OPERATIONAL PURPOSE
guests	id	None	Stores guest user profiles, contact metrics, and mailing records.
rooms	id	None	Maintains room inventory, pricing tiers, amenities, and availability status.
bookings	id	guest_id → guests.id room_id → rooms.id	Junction entity linking guests to rooms with date ranges and billing status.
reviews	id	None (Standalone)	Stores guest ratings and feedback; uses free-text identity strings.

3. ENTITY RELATIONSHIP DIAGRAM (ERD)



4. TABLE SCHEMAS & SAMPLE DATA

4.1 Table: `guests`

Stores information about individual accounts making reservations at the hotel.

COLUMN	TYPE	KEY	DESCRIPTION	ID	NAME	EMAIL	PHONE	ADDRESS
id	int	PK	Unique guest ID	1	Maria Lopez	maria.lopez@email.com	555-201-3344	12 Oak St, Austin, TX
name	string		Full legal name	2	James Chen	jchen@email.com	555-874-1290	450 Pine Ave, Seattle, WA
email	string		Contact email	3	Aisha Patel	aisha.p@email.com	555-639-7781	88 Elm Rd, Boston, MA
phone	string		Phone number					
address	string		Mailing address					

4.2 Table: `rooms`

Stores physical inventory, pricing tiers, amenities, and real-time occupancy status.

COLUMN	TYPE	KEY	DESCRIPTION	ID	TYPE	PRICE	CAPACITY	STATUS
id	int	PK	Unique room ID	101	Standard	129.00	2	Available
room_type	string		Category name	205	Deluxe	189.00	2	Occupied
description	string		Room features	310	Suite	349.00	4	Available
price	decimal		Nightly rate (\$)					
amenities	string		Feature list					
capacity	int		Max occupancy					
availability_status	string		Current state					

4.3 Table: `bookings`

Records room reservations, associating guests with rooms for specified stay periods.

ID	GUEST_ID	ROOM_ID	CHECKIN	CHECKOUT	BOOKING_STATUS	PAYMENT_STATUS	DATE_CREATED
1001	1 (FK)	205 (FK)	2025-03-10	2025-03-13	Confirmed	Paid	2025-02-15 09:22:10
1002	2 (FK)	101 (FK)	2025-03-12	2025-03-14	Confirmed	Pending	2025-02-20 14:05:31
1003	3 (FK)	310 (FK)	2025-02-01	2025-02-05	Completed	Paid	2025-01-10 18:47:02

4.4 Table: `reviews`

Stores feedback and approval state submitted by staying guests.

ID	REVIEWER_INFO	RATING	REVIEW_TEXT	SUBMISSION_DATE	APPROVAL_STATUS
----	---------------	--------	-------------	-----------------	-----------------